

CSCI 361: Exercise # 9

Your Name: _____

Due: Tuesday, Oct 21, in class

Assigned reading for this lecture:

- Sipser Third Edition Chap 4.1 (Decidability)

Questions:

1. Use equivalence to a multi-tape Turing machine to justify why a TM can make "subroutines calls" (or functions call) such as NFA to DFA conversion or calling another Turing machine as described in Theorems 4.2-4.5.
2. Why can't we prove that the language EQ_{CFG} is decidable by doing a similar argument for CFGs as Theorem 4.5 does for regular languages?